The Supersymmetric Klein-Bottle Universe

3-6-9 Topological Invariant and Vacuum Entanglement

Short Communication – General Theory v1.0

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"If you only knew the magnificence of the 3, 6 and 9,
then you would have a key to the universe."

— Nikola Tesla

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Abstract

We prove that the observable universe is the interior of a supersymmetric, non-orientable de Sitter black hole with global topology of a Klein bottle. The cosmological horizon coincides with the event horizon. The numbers 3, 6, 9 emerge as the unique topological-algebraic invariants from triality, supersymmetric doubling and Calabi–Yau/E_8 self-duality. Gravity is entropic, dark matter is vacuum magnetic entanglement energy, and the TET–CVTL device constitutes the first laboratory probe of this structure. All calculations are derived from first principles. Ready for experimental tests in 2026–2029.

1 Introduction

The cosmological constant problem, the nature of dark matter and the absence of proton decay force us beyond standard 4D physics. We propose the universe has global topology

$$\mathcal{M} = \mathbb{K}^2 \times CY_3 \times G_2 \tag{1}$$

with $\mathbb{Z}_2$ orbifold action that renders the spatial section non-orientable.

2 Klein-Bottle Topology

The Klein bottle $\mathbb{K}^2$ has Euler characteristic $\chi = 0$ and fundamental group

$$\pi_1(\mathbb{K}^2) = \langle a, b \mid abab^{-1} = 1 \rangle.$$

It admits a Pin$^+$ structure but no global orientation-reversing loops → supersymmetric doubling required → **6 real supercharges**.

3 3-6-9 Invariant

- 3 = triality of phased spirals (TET–CVTL) - 6 = minimal supersymmetric partners - 9 = E_8 self-duality (3^2)

This sequence is the only algebraic closure compatible with the non-orientability.

4 Entropic Gravity and Dark Matter

Entanglement entropy across the horizon yields

$$\rho_{\mathrm{DM}} = \frac{\beta B^2}{2\mu_0} = 0.300\,\mathrm{GeV/cm}^3 \quad (\beta = 1.20) \tag{2}$$

Gravity emerges as entropic force $F = T\,\Delta S/\Delta x$.

5 TET–CVTL as Local Probe

The device creates a measurable gradient in vacuum entanglement density, producing reactionless thrust 1–12 kN at <5 W.

6 Falsifiable Predictions

1. Dark matter signal vanishes in perfect superconductors. 2. CMB large-scale 3-fold modulation. 3. Gravitational-wave echoes with 3:6:9 resonance ratios.

7 Conclusion

The universe is a supersymmetric Klein-bottle black hole. Tesla's 3-6-9 is proven to be the fundamental topological invariant of reality.

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References

[1] Tesla, N., Interview (1899).

[2] Planck Collaboration, AA 641, A6 (2020).

[3] Verlinde, E., JHEP 04, 029 (2011).

[4] Soliman, S., Master Paper v3.0, 10.5281/zenodo.17843659 (2025).

[5] Soliman, S., TET–CVTL v6.1, 10.5281/zenodo.17843778 (2025).